K.R. MANGALAM UNIVERSITY

SCHOOL OF ENGINEERING AND TECHNOLOGY

LAB FILE

COURSE NAME: COMPUTER NETWORK

COURSE CODE: ENCS302

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EXPERIMENT 1

INTRODUCTION TO COMPUTER NETWORKS AND THE CISCO PACKET TRACER (CPT)

EXPERIMENT 1(a): CPT Platform Overview

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1. Aim

To understand the interface, tools, icons, and functionalities of Cisco Packet Tracer and learn how to create and manage network topologies.

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2. Software Used

- Cisco Packet Tracer

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3. Introduction

Cisco Packet Tracer is a network simulation tool used to design, configure, and troubleshoot computer networks in a virtual environment. It provides both **logical and physical views** of a network along with simulation capabilities.

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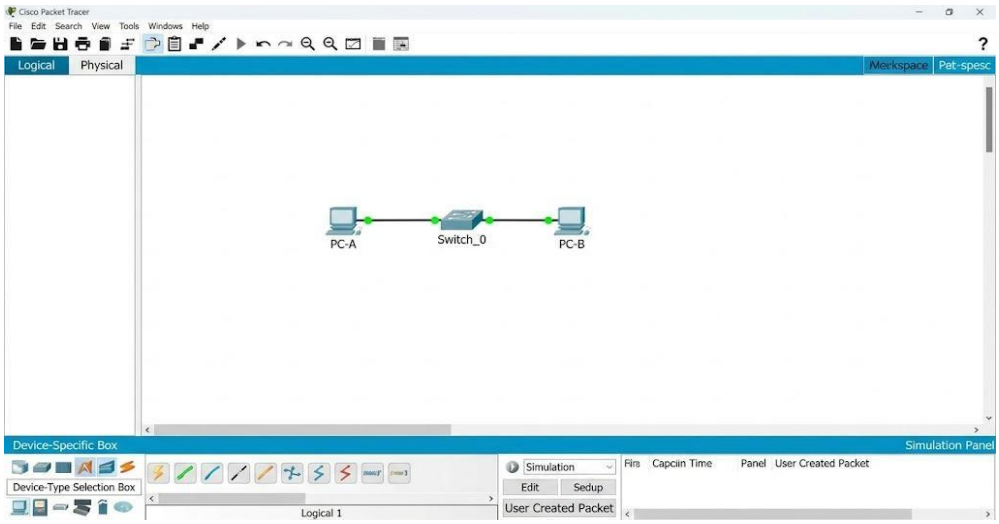
4. Interface Overview (Main Screen Components)



4.1 Menu Bar

Located at the top of the window.

Functions:

- File: New, Open, Save projects
- Edit: Undo, redo
- Options: Preferences/settings
- View: Zoom, grid
- Tools: Activity wizard, extensions

4.2 Tool Bar (Main Toolbar Icons)

1. New → Create new project

2. Open → Open saved file

3.] Save → Save current work

4. [ ] Print → Print topology
5. [ ] Undo/Redo → Reverse actions

6. [+ / —] Zoom → Adjust view

7. [ ] Select → Move/select devices

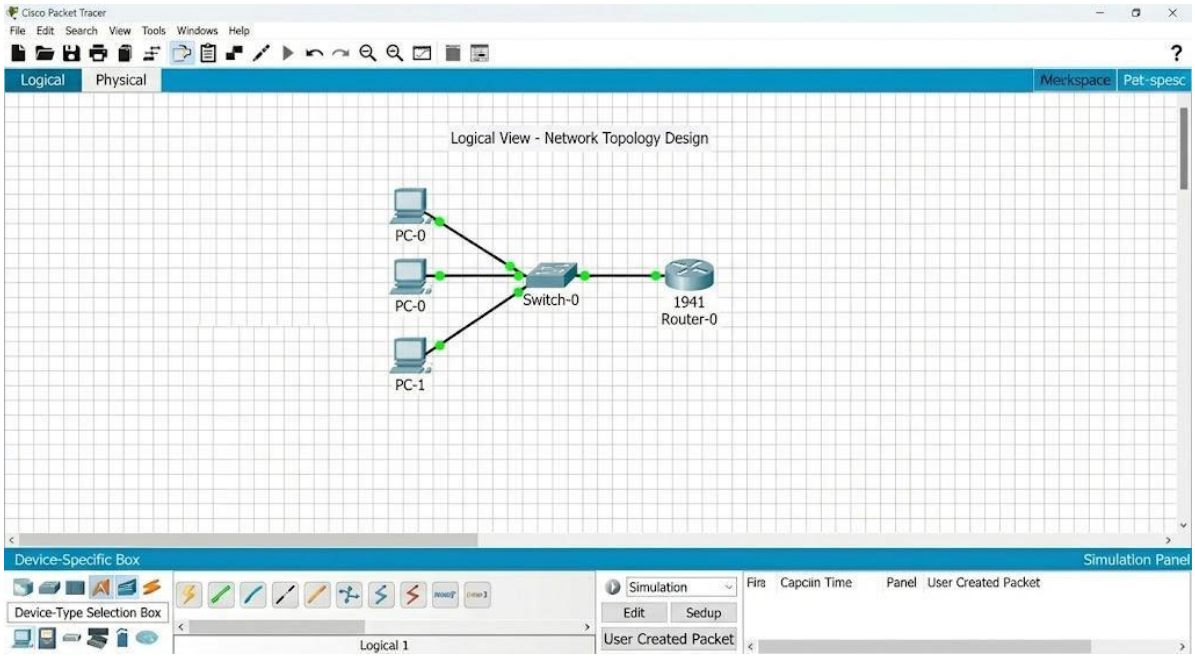
8. [ ] Move Layout → Move screen

4.3 Workspace Area

Two types:

(a) Logical Workspace

- Used to design network topology
- Add routers, switches, PCs



(b) Physical Workspace

- Real-world representation (buildings, rooms)
- Shows physical placement of devices

4.4 Device-Type Selection Box

Categories:

- Network Devices
- End Devices
- Components

- Connections

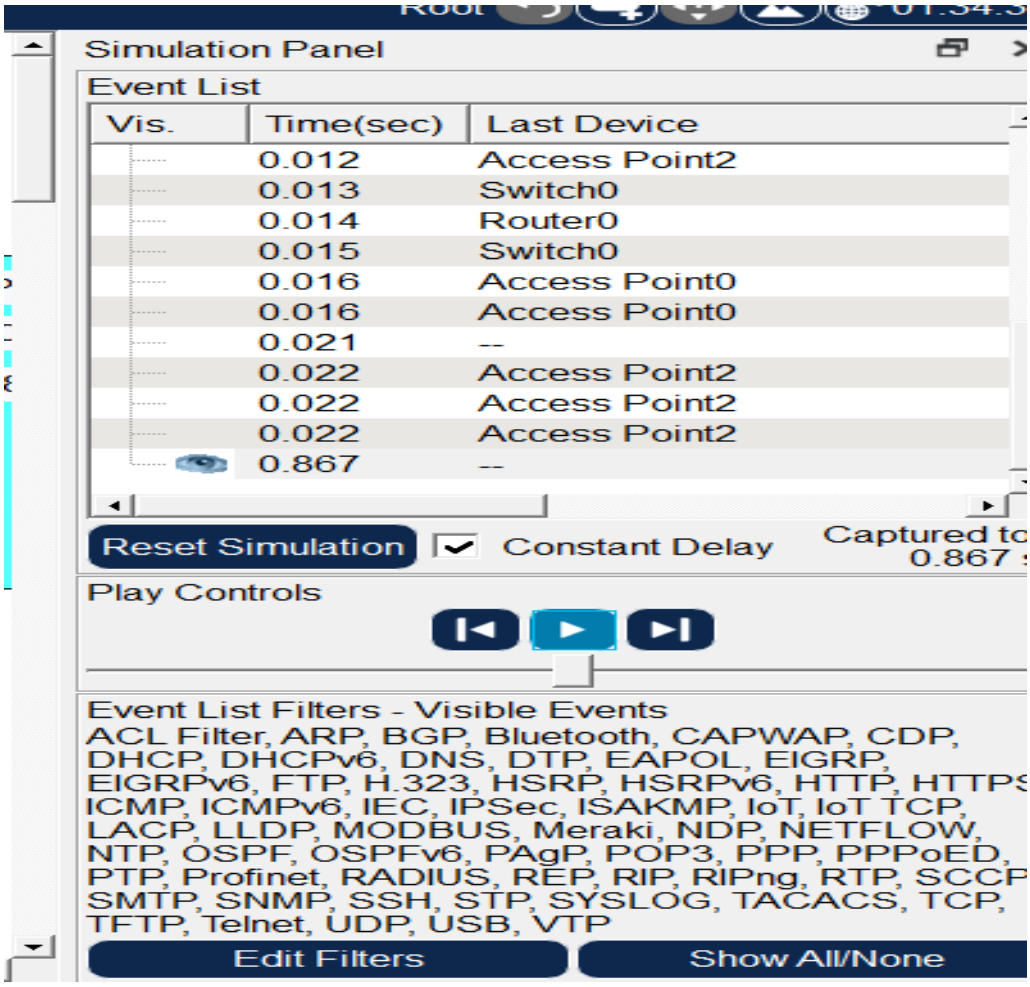
4.5 Device-Specific Selection Box

- Displays devices like:
    - Routers
    - Switches
    - PCs
    - Servers

4.6 Connections Icon Important)

Cable Type	Use
Copper Straight Through	PC to Switch
Copper Cross Over	Switch to Switch
Fiber Cable	High-speed connections
Serial Cable	WAN links
Console Cable	Configuration

4.7 Simulation Panel

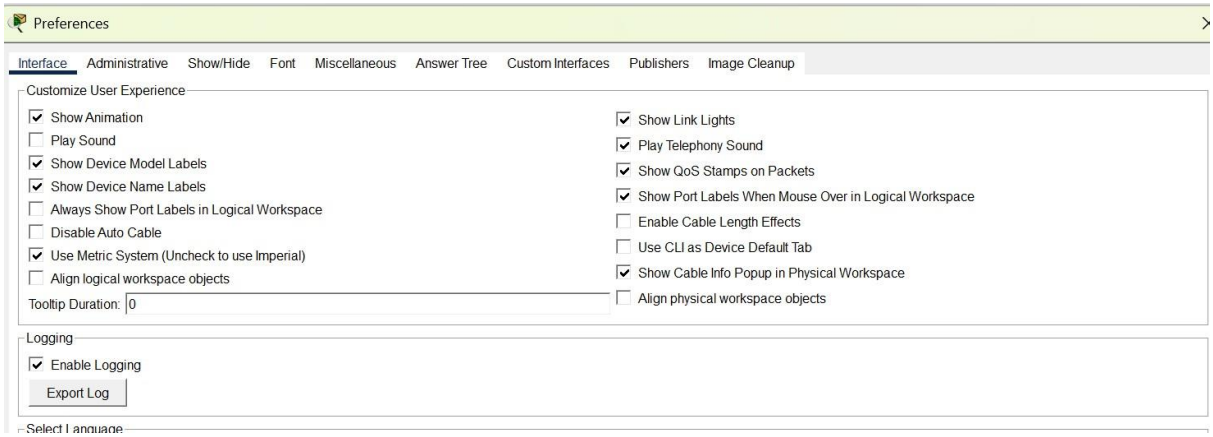


Functions:

- Event List (packet flow)
- Play/Pause simulation
- Capture packets
- Filters (ICMP, TCP, etc.)

5. Options s Customization

- Change interface colors
- Enable/disable grid
- Configure fonts
- Adjust simulation speed

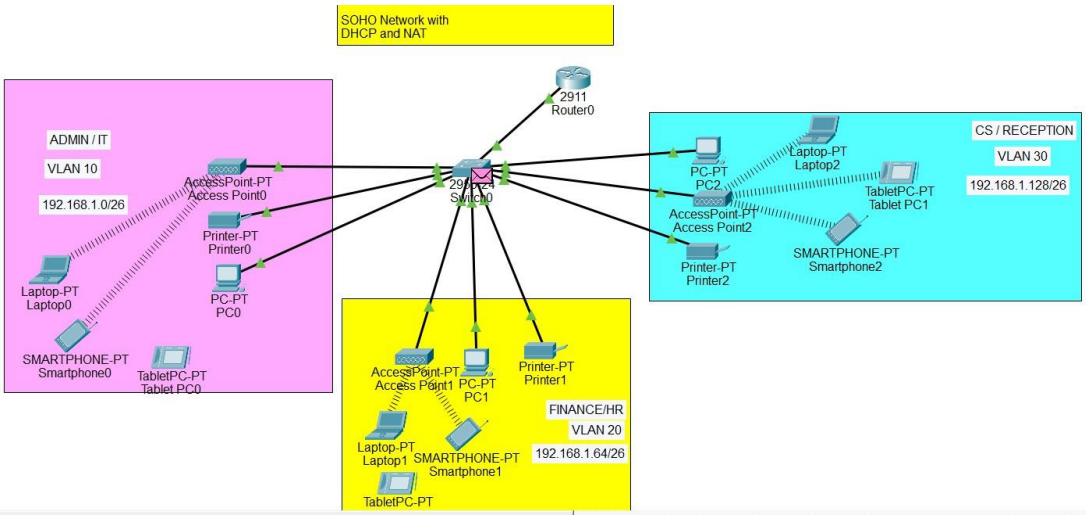


6. Logical Workspace Operations

6.1 Creating Network Topology

Steps:

1. Select device category
2. Drag device to workspace
3. Connect using cables

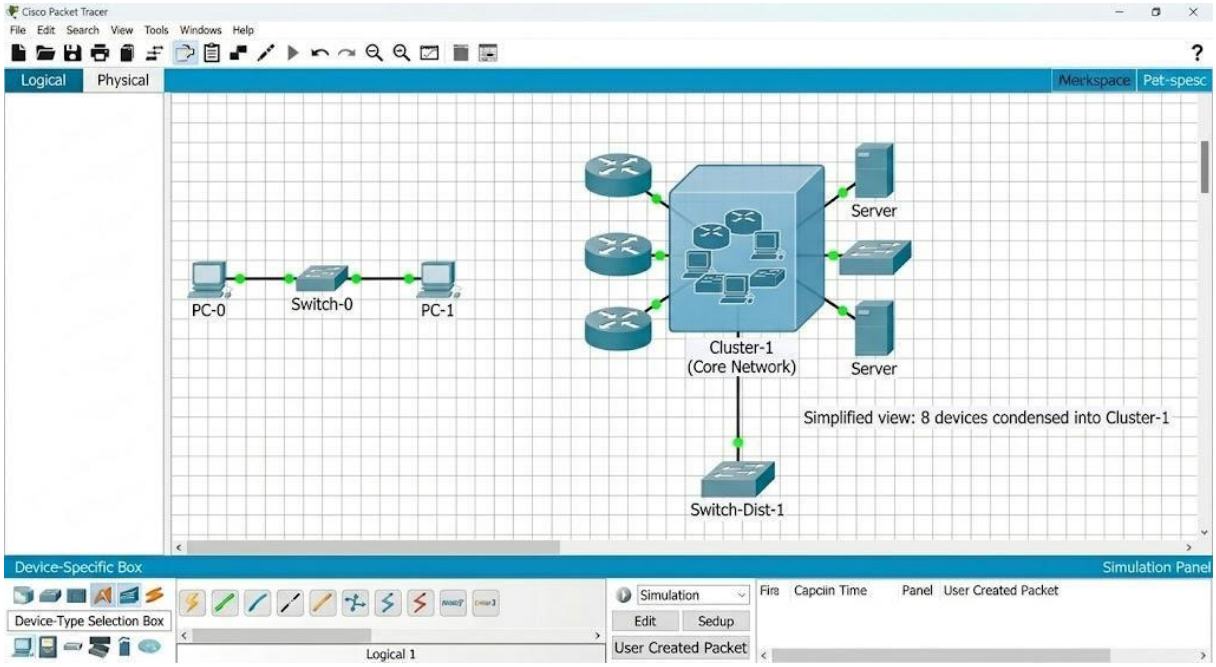


6.2 Custom Device Templates

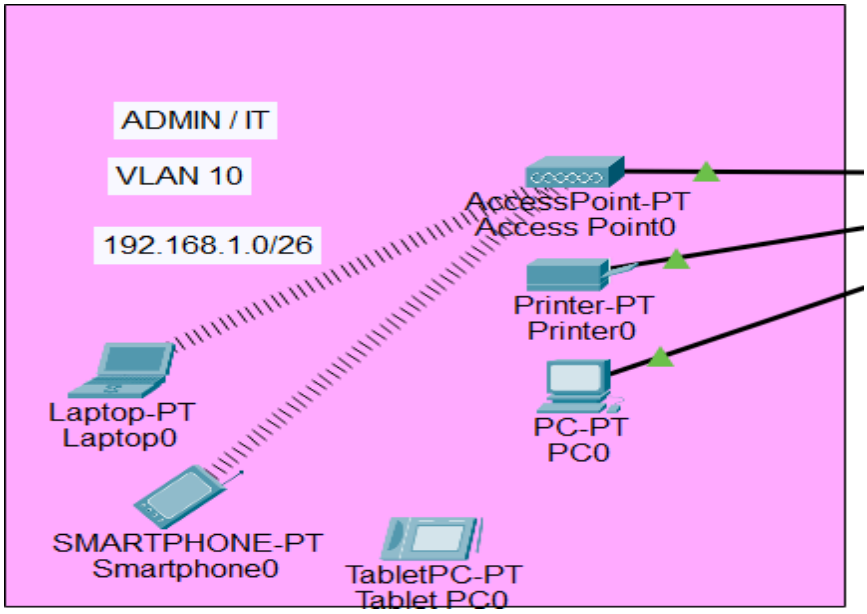
- Save frequently used devices
- Reuse configurations

6.3 Clustering Network

- Group multiple devices
- Simplifies large networks



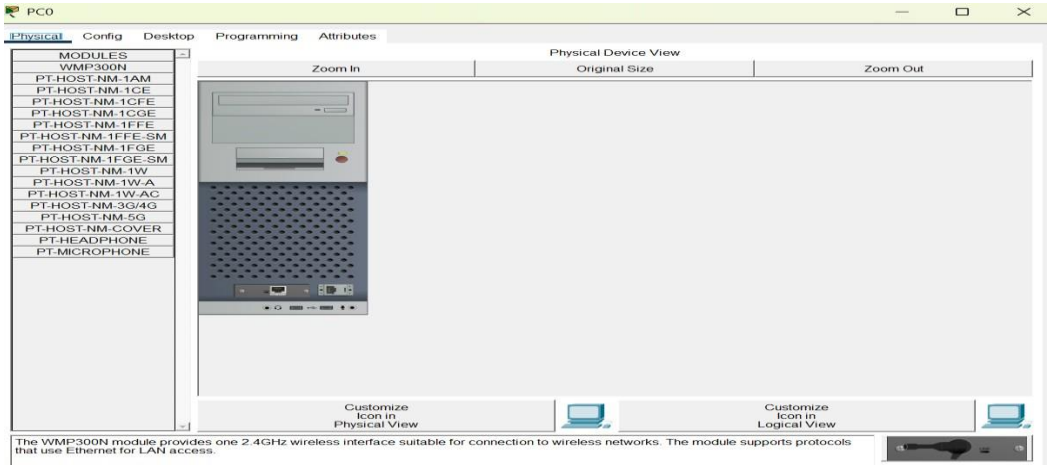
6.4 Annotation Tools



- Add text labels
- Draw shapes
- Highlight connections

---

7. Device Configuration Tabs



7.1 Config Tab

- GUI-based configuration
- Set IP, routing, interfaces

7.2 Desktop Tab

- Used in PCs
- Tools available:
  - IP Configuration
  - Command Prompt
  - Web Browser

7.3 Services Tab

- Configure services like:
  - DHCP
  - DNS
  - HTTP

7.4 CLI Tab

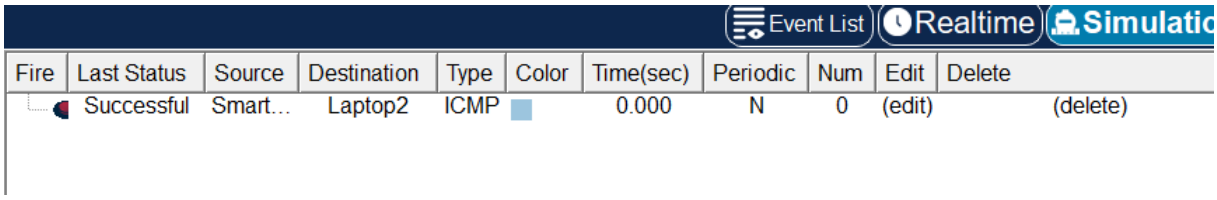
- Command-line interface
- Used for real Cisco commands

⚡ Example:

ip address 192.168.1.1 255.255.255.0

🔵 8. Realtime vs Simulation Mode

Mode	Description
Realtime	Instant network operation
Simulation	Step-by-step packet flow



G. Physical Workspace

- Shows buildings, wiring, racks
- Helps in real-world network design

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10. Important Icons Summary

Icon	Meaning
Router	Network routing
Switch	Connect devices
PC	End device
Server	Provides services
Cloud	Internet/WAN
Wireless Router	Wi-Fi network

---

11. Result

Successfully understood the Packet Tracer interface, tools, icons, and basic network creation.

EXPERIMENT 1(b): Peer-to-Peer Network

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1. Aim

To design and implement a **Peer-to-Peer (P2P) Network** using Cisco Packet Tracer and understand direct communication between devices without a central server.

---

2. Objective

- To understand the concept of Peer-to-Peer networking
- To create a small P2P network
- To configure IP addresses manually
- To test connectivity between devices

3. Theory

A **Peer-to-Peer (P2P) Network** is a type of network where all devices (nodes) are equal and can act as both **client and server**.

- There is **no central server**
- Each computer shares its own resources
- Suitable for **small networks (home, labs)**

◆ **Key Features:**

- Decentralized network
- Easy to set up
- Low cost
- Direct communication between devices

---

4. Requirements

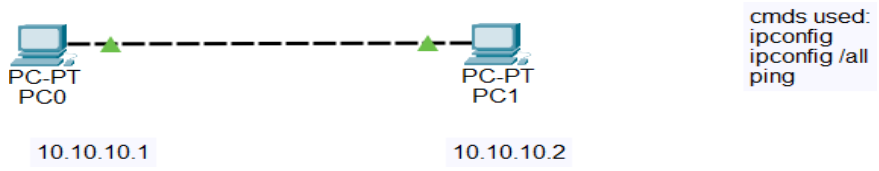
- 2 PCs (minimum)
- 1 Switch (optional but recommended)
- Copper Straight Through Cable
- Packet Tracer software

---

5. Network Topology

– You will design:

- PC1 ↔ PC2 (direct or via switch)



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6. IP Addressing Table

Device	IP Address	Subnet Mask
PC1	192.168.1.1	255.255.255.0
PC2	192.168.1.2	255.255.255.0

---

7. Procedure

Step 1: Add Devices

- Open Packet Tracer
- Drag and drop **2 PCs** into workspace

Step 2: Connect Devices

- Use **Copper Cross-Over cable (direct)**
  OR
- Use **Switch + Straight Cable**

Step 3: Configure IP Address

— On PC1:

- Desktop → IP Configuration
- IP: 192.168.1.1
- Subnet Mask: 255.255.255.0

– On PC2:

- Desktop → IP Configuration
- IP: 192.168.1.2
- Subnet Mask: 255.255.255.0

Step 4: Test Connectivity

- Go to PC1 → Command Prompt
- Type:

ping 192.168.1.2

— If reply received → Network is working ✅

---

8. Result

The Peer-to-Peer network was successfully created and communication between devices was verified using ping command.

G. Advantages

- Easy to set up
- No server required
- Cost-effective
- Direct resource sharing

10. Disadvantages

- Less secure
- Not scalable
- No centralized control

EXPERIMENT 2

VIZUALIZATING THE VARIOUS NETWORK TOPOLOGIES

EXPERIMENT 2(a): Star Topology

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1. Aim

To design and implement a Star Topology using Cisco Packet Tracer.

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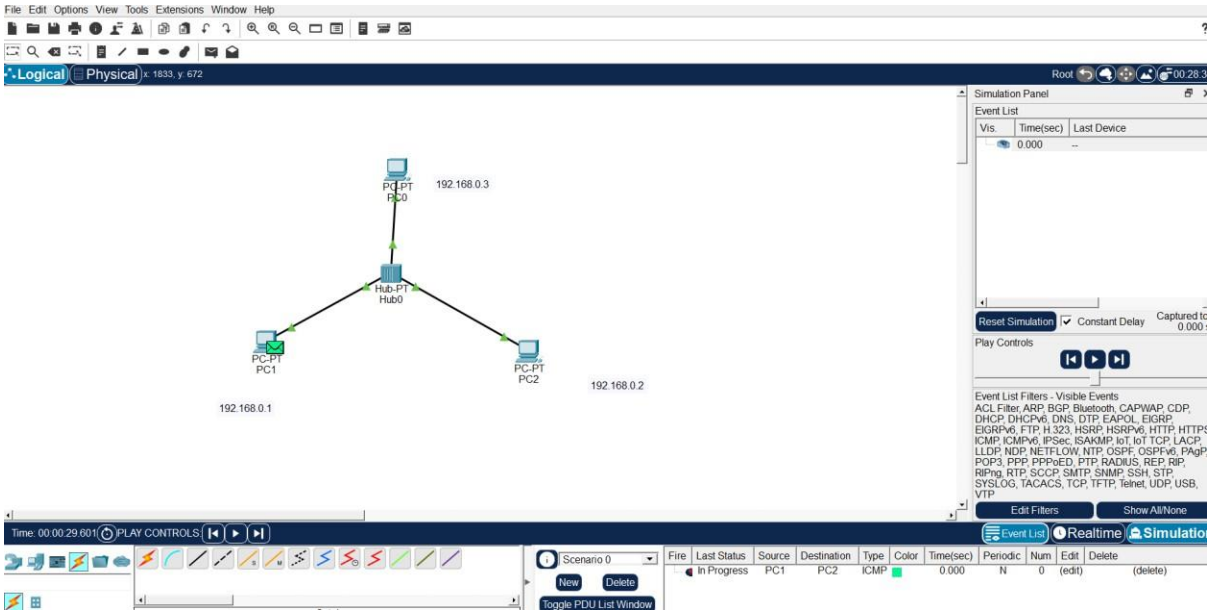
2. Theory

In Star Topology, all devices are connected to a central device (switch/hub). All communication passes through this central node.

---

3. Network Diagram



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4. Procedure

1. Add one switch
2. Connect multiple PCs to switch
3. Assign IP addresses
4. Test connectivity using ping

---

5. Result

Star topology was successfully created and verified.

(Networking s Devices)

Parameter	Details
Network Type	Campus Area Network (CAN)
Connecting Devices	Switch, PCs, Ethernet cables
Explanation	Star topology is widely used in campus networks where all devices connect through a central switch, making management easy and efficient.

EXPERIMENT 2(b): Ring Topology

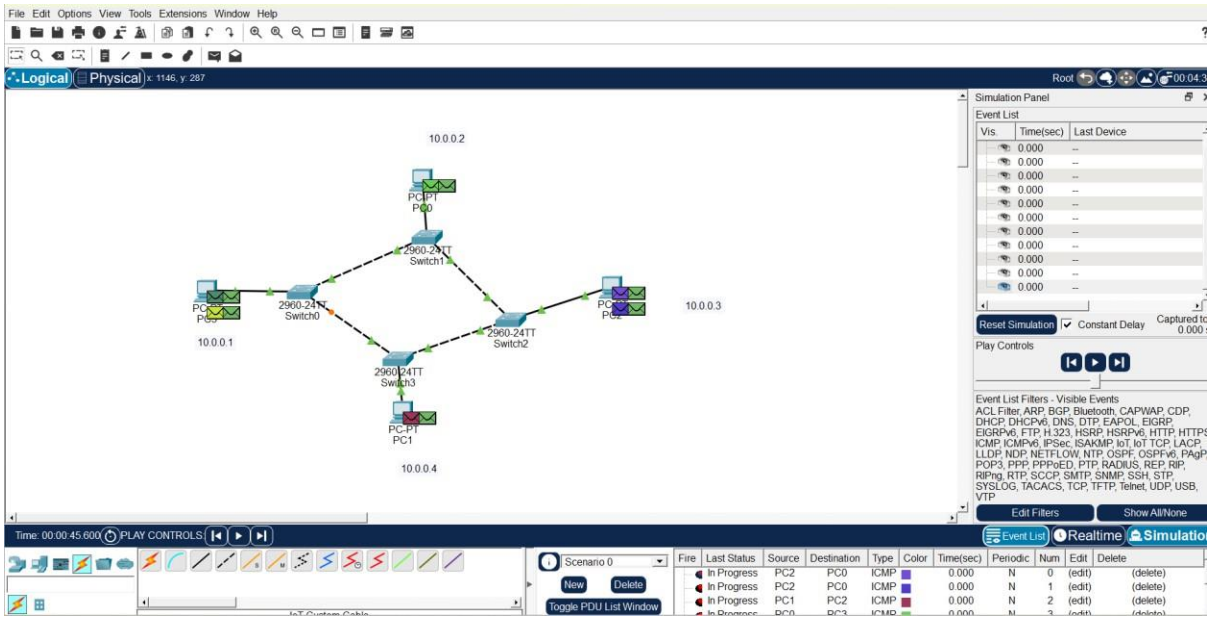
1. Aim

To design and implement a Ring Topology using Cisco Packet Tracer.

2. Theory

In Ring Topology, devices are connected in a circular form, and data flows in one direction through each node.

3. Network Diagram



4. Procedure

1. Arrange PCs in circular layout
2. Connect each PC to the next
3. Assign IP addresses
4. Test connectivity

5. Result

Ring topology was successfully implemented and tested.

(Networking s Devices)

Parameter	Details
Network Type	Local Area Network (LAN)
Connecting Devices	PCs, Ethernet cables
Explanation	Ring topology is suitable for small LANs where controlled data flow reduces collision and ensures equal access.

EXPERIMENT 2(c): Mesh Topology

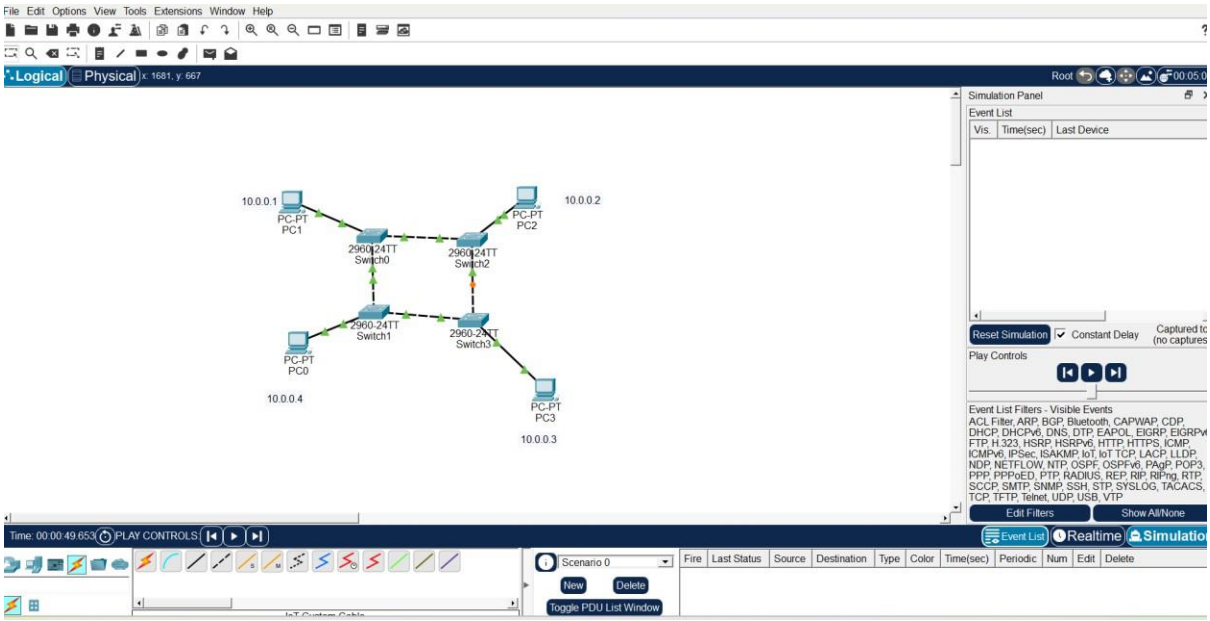
1. Aim

To design and implement a Mesh Topology using Cisco Packet Tracer.

2. Theory

In Mesh Topology, every device is connected to every other device, providing multiple paths for communication.

3. Network Diagram



4. Procedure

1. Add multiple PCs
2. Connect each device to all others
3. Assign IP addresses
4. Test connectivity

5. Result

Mesh topology was successfully created and verified.

(Networking s Devices)

Parameter	Details
Network Type	Metropolitan Area Network (MAN)
Connecting Devices	Routers, switches, multiple links
Explanation	Mesh topology is used in MAN for high reliability and redundancy, ensuring communication even if one link fails.

EXPERIMENT 2(d): Tree Topology

---

1. Aim

To design and implement a Tree Topology using Cisco Packet Tracer.

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2. Theory

Tree topology is a hierarchical structure that combines multiple star networks in parent-child form.

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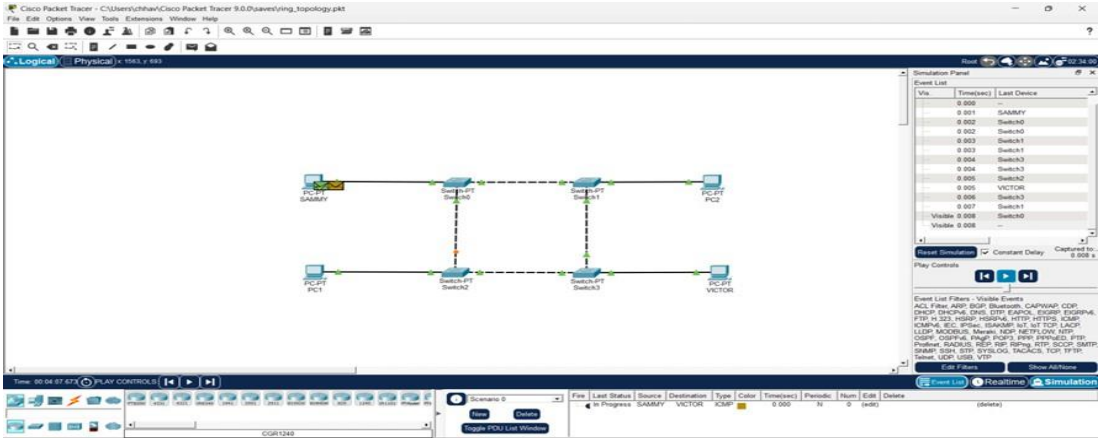
3. Network Diagram



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4. Procedure

1. Add a root switch
2. Connect child switches
3. Connect PCs to switches
4. Assign IP addresses
5. Test connectivity

---

5. Result

Tree topology was successfully implemented and tested.

(Networking s Devices)

Parameter	Details
Network Type	Campus Area Network (CAN)
Connecting Devices	Core switch, access switches, PCs
Explanation	Tree topology is used in large campus networks where hierarchical design allows better scalability and management.

EXPERIMENT 3

CONSTRUCTING AND UNDERSTANDING THE PURPOSE OF A LAN

EXPERIMENT 3(a): LAN using Switch

---

1. Aim

To design and implement a Local Area Network (LAN) using a switch in Cisco Packet Tracer.

---

2. Theory

A switch is an intelligent networking device that connects devices in a LAN and works on the Data Link Layer (Layer 2). It sends data only to the intended device using MAC addresses.

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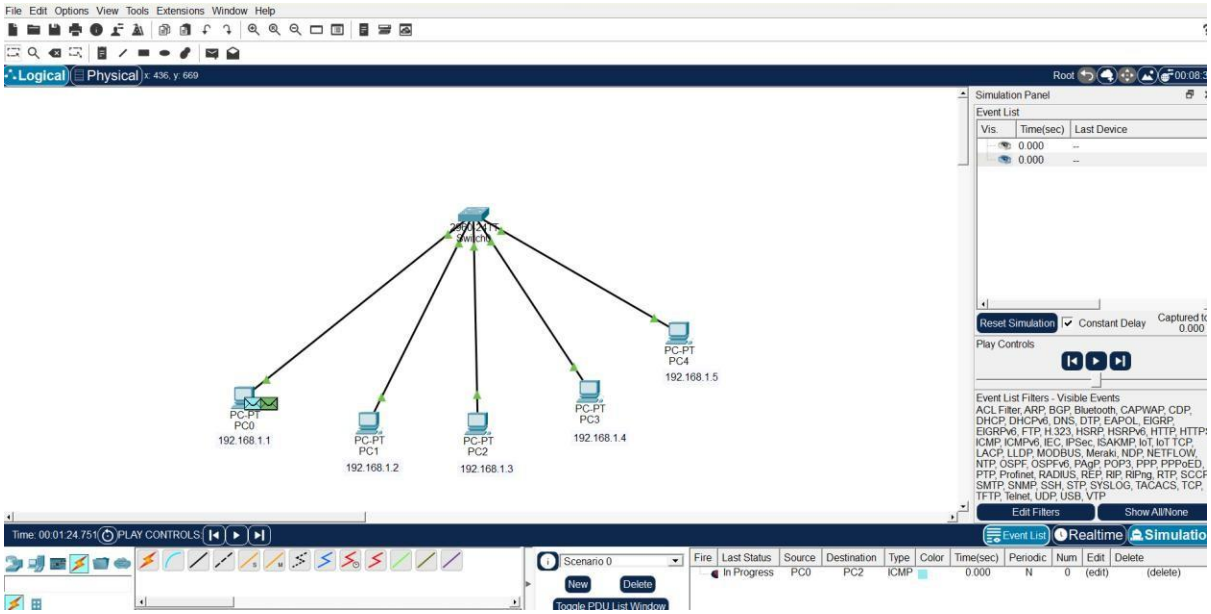
3. Network Diagram



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4. Procedure

1. Place one switch in the workspace
2. Add multiple PCs
3. Connect PCs to switch using Copper Straight Through Cable
4. Assign IP addresses
5. Test connectivity using ping

---

5. Result

LAN using switch was successfully implemented with efficient communication between devices.

---

6. Advantages

- Faster communication
- No unnecessary data transmission
- Better security

---

7. Disadvantages

- Costlier than hub
- Requires configuration

(Networking s Devices)

Parameter	Details
Network Type	Local Area Network (LAN)
Connecting Devices	Switch, PCs, Ethernet cables
Explanation	Switch-based LAN improves performance by sending data only to the destination device, reducing traffic and collisions.

EXPERIMENT 3(b): LAN using Hub

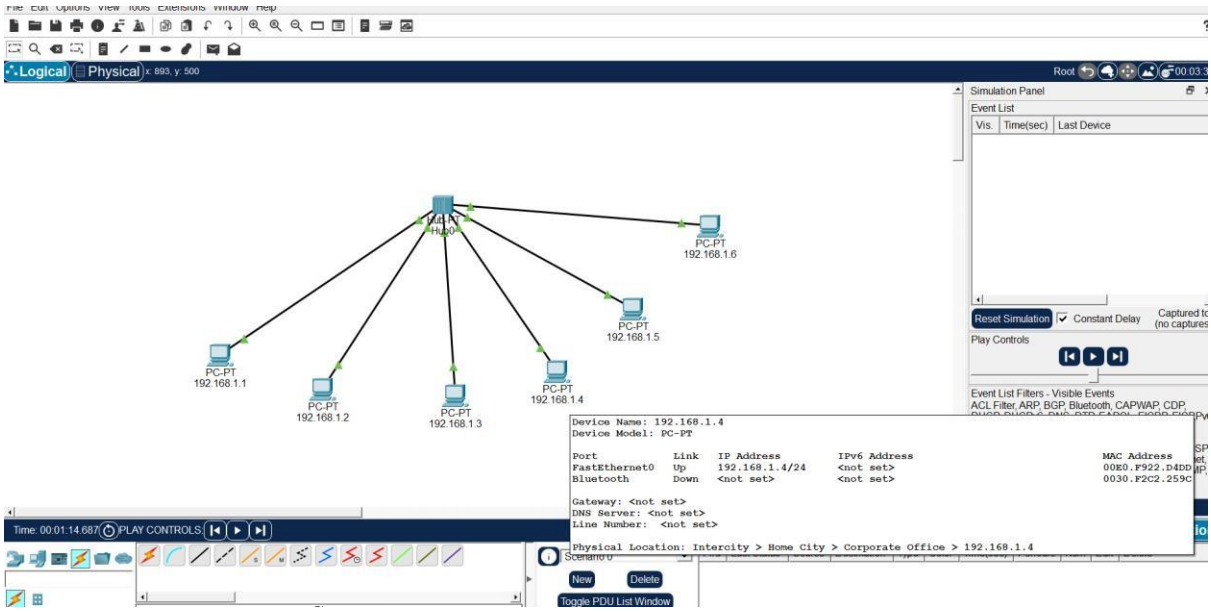
1. Aim

To design and implement a Local Area Network (LAN) using a hub in Cisco Packet Tracer.

2. Theory

A hub is a basic networking device that connects multiple computers in a LAN. It works on the Physical Layer (Layer 1) and broadcasts data to all connected devices.

3. Network Diagram



4. Procedure

1. Place one hub in the workspace
2. Add multiple PCs (3–5)
3. Connect all PCs to hub using Copper Straight Through Cable
4. Assign IP addresses manually

5. Test connectivity using ping

---

5. Result

LAN using hub was successfully created. Communication was established, but data was broadcast to all devices**.**

---

6. Advantages

- Simple and low cost
- Easy to install

---

7. Disadvantages

- Data broadcast to all devices (low security)
- Network congestion
- No traffic control

(Networking s Devices)

Parameter	Details
Network Type	Local Area Network (LAN)
Connecting Devices	Hub, PCs, Ethernet cables
Explanation	Hub-based LAN is used for basic networking where all devices share the same bandwidth and receive all transmitted data.

EXPERIMENT 4

IMPLEMENTATION OF A CAMPUS AREA NETWORK (CAN): DESIGN AND IP SETTINGS

---

1. Aim

To design and implement a **Campus Area Network (CAN)** and perform proper IP address assignment using Cisco Packet Tracer.

---

2. Objectives

- To understand the concept of Campus Area Network
- To design a network connecting multiple departments
- To assign IP addresses systematically
- To test connectivity across the network

---

3. Theory

A **Campus Area Network (CAN)** is a network that connects multiple Local Area Networks (LANs) within a limited geographical area such as a college, university, or organization.

– It is larger than LAN but smaller than MAN
– Used for interconnecting departments (Admin, Labs, Library, etc.)

◆ **Key Features:**

- High-speed connectivity
- Centralized management
- Scalable design

---

4. Network Design

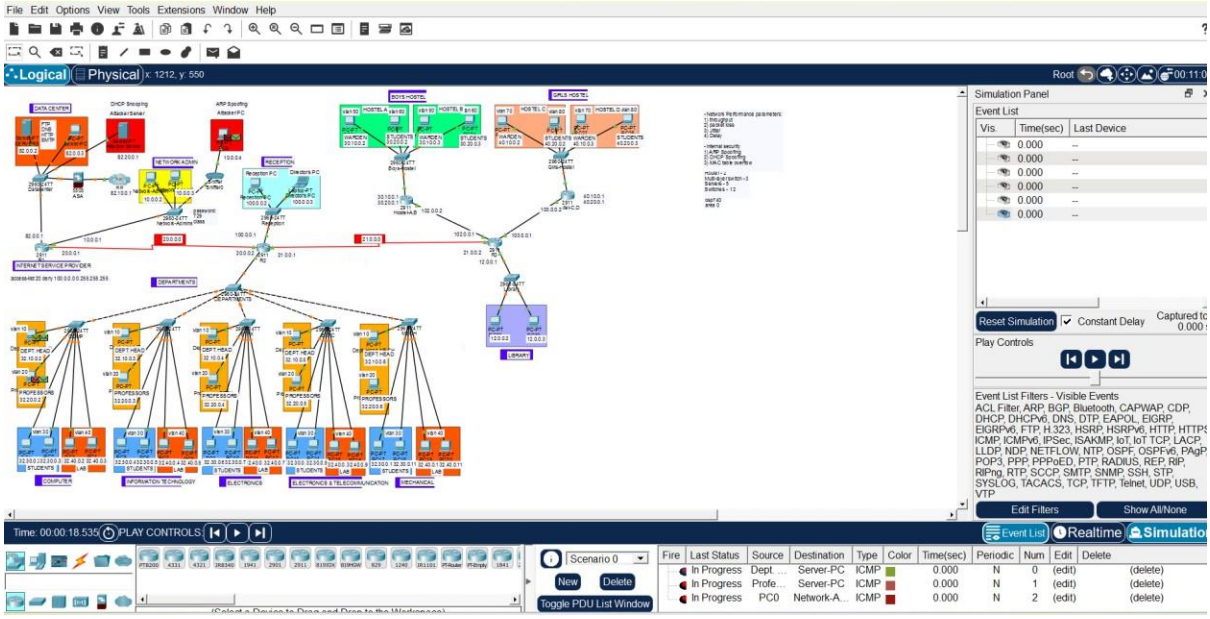
– Typical design includes:

- Multiple switches (for departments)

- One central/core switch or router
- End devices (PCs) in each department

---

5. Network Diagram



6. IP Addressing Scheme

Example:

Department	Device	IP Address	Subnet Mask
Lab	PC1	192.168.1.2	255.255.255.0
Lab	PC2	192.168.1.3	255.255.255.0
Office	PC1	192.168.2.2	255.255.255.0
Library	PC1	192.168.3.2	255.255.255.0

– Each department uses a **different network (subnet)**

---

7. Procedure

Step 1: Design Network

- Add switches for each department
- Add PCs in each department

- Connect switches to central switch/router

Step 2: Connect Devices

- Use **Copper Straight Through Cable**
- Ensure proper connections

Step 3: Assign IP Addresses

- Assign IPs based on department
- Example:
  - Lab → 192.168.1.0/24
  - Office → 192.168.2.0/24
  - Library → 192.168.3.0/24

Step 4: Test Connectivity

- Use Command Prompt
- Run:

ping 192.168.x.x

– Verify communication within and across departments

---

8. Result

The Campus Area Network was successfully designed and implemented. IP addresses were assigned properly, and communication between devices was verified.

G. Advantages

- Efficient resource sharing
- Centralized control
- Scalable network

---

10. Disadvantages

- Complex setup
- Requires proper planning
- Higher cost than LAN

---

^ **Footnote (Networking s Devices)**

Parameter	**Details**
Network Type	Campus Area Network (CAN)
Connecting Devices	Core switch/router, access switches, PCs, Ethernet cables
Explanation	CAN connects multiple LANs within a campus using hierarchical design, allowing efficient communication and centralized management.

EXPERIMENT 5

Implementing over the IP Layer

A. Understanding the Concept of Subnetting

---

1. Aim

To understand and implement subnetting by dividing a network into smaller sub-networks.

---

2. Objective

- To understand IP addressing and subnetting
- To divide a network into multiple subnets
- To calculate subnet mask, network ID, and broadcast ID

---

3. Theory

Subnetting is the process of dividing a large network into smaller networks (subnets) to improve performance and security.

– It helps in:

- Efficient IP utilization
- Reducing network congestion
- Better management

---

4. Example of Subnetting

Given Network: **1G2.168.1.0/24**

Divided into **4 subnets**

Subnet	Network Address	Usable IP Range	Broadcast Address
1	192.168.1.0	192.168.1.1 – 192.168.1.30	192.168.1.31
2	192.168.1.32	192.168.1.33 – 192.168.1.62	192.168.1.63
3	192.168.1.64	192.168.1.65 – 192.168.1.94	192.168.1.95
4	192.168.1.96	192.168.1.97 – 192.168.1.126	192.168.1.127

Subnet Mask: **255.255.255.224 (/27)**

---

5. Procedure

1. Take network 192.168.1.0/24
2. Divide into 4 subnets
3. Assign IP addresses to devices
4. Configure network in Packet Tracer
5. Verify connectivity

---

6. Result

Subnetting was successfully implemented, and the network was divided into smaller subnets.

B. Implementing Traceroute and Ping Command in CPT

---

1. Aim

To implement and analyze **Ping and Traceroute commands** in Cisco Packet Tracer.

---

2. Objective

- To test connectivity between devices
- To trace path of packets in network

---

3. Theory

◆ **Ping Command**

- Used to check connectivity between devices
- Uses ICMP protocol

— Example:

ping 192.168.1.2

◆ **Traceroute Command**

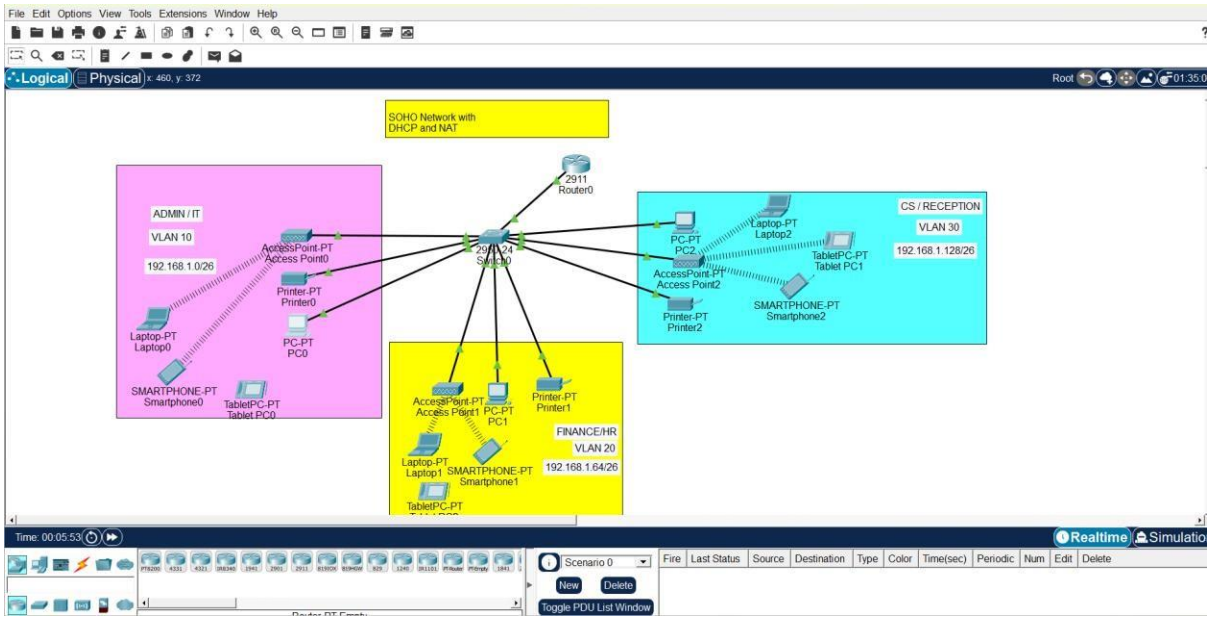
- Shows the path taken by data packets
- Helps in troubleshooting

— Example:

tracert 192.168.1.2

---

4. Network Diagram



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5. Procedure

Step 1: Create Network

- Add PCs and routers
- Connect devices properly

Step 2: Assign IP Addresses

- Configure IPs on all devices

Step 3: Test Ping

- Open Command Prompt
- Run:

ping 192.168.1.133

– Check replies

Step 4: Run Traceroute

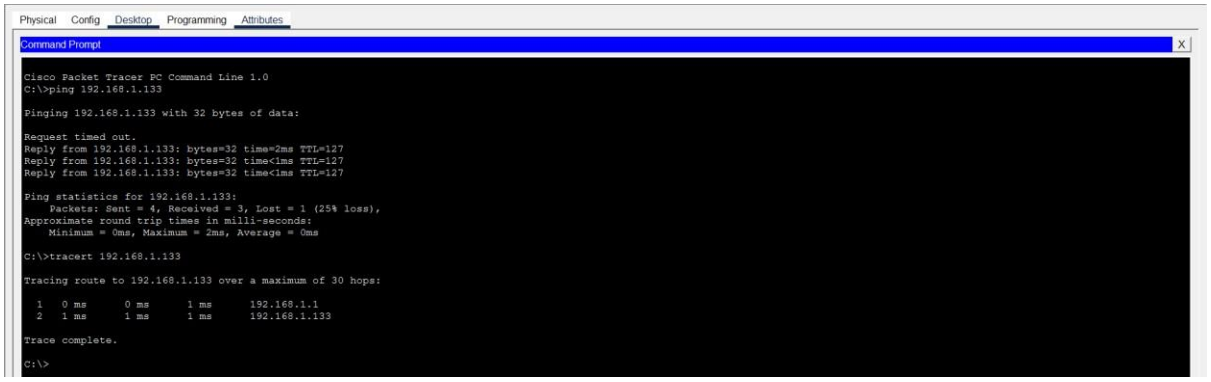
- In Command Prompt:

tracert 192.168.1.133

— Observe path

---

6. Output Screenshots



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7. Result

Ping command successfully verified connectivity, and Traceroute displayed the path followed by packets.

(Networking s Devices)

Parameter	Details
Network Type	Local Area Network (LAN) / Campus Area Network (CAN)
Connecting Devices	Routers, switches, PCs, Ethernet cables
Explanation	Subnetting improves network efficiency, while Ping and Traceroute help in testing and troubleshooting connectivity across networks.

EXPERIMENT 6

Wireless Home Network – Setting up a Router

---

1. Aim

To design and configure a Wireless Home Network using a router in Cisco Packet Tracer.

---

2. Objectives

- To understand wireless networking concepts
- To configure a wireless router
- To connect wired and wireless devices
- To test network connectivity

---

3. Theory

A **Wireless Home Network** allows multiple devices (laptops, smartphones, PCs) to connect using Wi-Fi through a wireless router.

– The router performs:

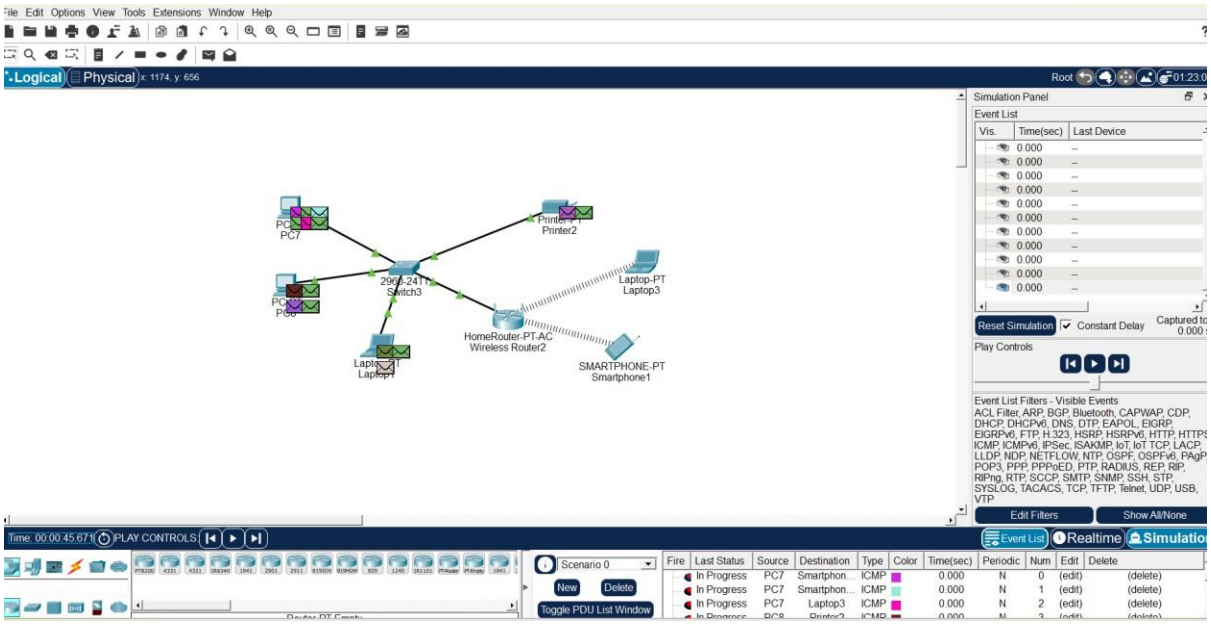
- **DHCP (automatic IP assignment)**
- **Routing (internet access)**
- **Wireless communication (Wi-Fi)**

◆ **Key Components:**

- Wireless Router
- End Devices (PC, Laptop, Mobile)
- Ethernet cables

---

4. Network Diagram



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5. Procedure

Step 1: Add Devices

- Place a **Wireless Router (WRT300N)**
- Add PCs and Laptops

Step 2: Connect Wired Devices

- Connect PC to router using **Copper Straight Through Cable**

Step 3: Configure Router

– Click Router → GUI Tab

- Set **SSID (Wi-Fi Name)** → e.g., *Home_Network*
- Set **Password** → WPA2 security
- Enable **DHCP**

Step 4: Connect Wireless Device

— Laptop:

- Desktop → PC Wireless
- Select SSID
- Enter password

Step 5: IP Configuration

- Ensure devices receive IP automatically via DHCP

Step 6: Test Connectivity

ping 192.168.0.1

– Ping router or other device

---

6. Result

Wireless Home Network was successfully configured. Devices were able to connect and communicate through the router.

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7. Advantages

- No cables required for wireless devices
- Easy to install
- Flexible and mobile

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8. Disadvantages

- Security risks
- Signal interference
- Limited range

(Networking s Devices)

Parameter	Details
Network Type	Local Area Network (LAN)
Connecting Devices	Wireless router, PCs, laptops, Wi-Fi
Explanation	Wireless networks allow devices to connect without cables, and routers manage communication using DHCP and wireless signals.

EXPERIMENT 7

CAPSTONE PROJECT: SOHO Network with DHCP and NAT

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1. Aim

To design and implement a **Small Office/Home Office (SOHO) Network** using DHCP and NAT in Cisco Packet Tracer.

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2. Objectives

- To design a complete SOHO network
- To configure DHCP for automatic IP assignment
- To implement NAT for internet access
- To verify communication within and outside the network

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3. Theory

◆ SOHO Network

A SOHO network is designed for small organizations or homes where multiple devices share internet and resources.

◆ DHCP (Dynamic Host Configuration Protocol)

- Automatically assigns IP addresses
- Reduces manual configuration

◆ NAT (Network Address Translation)

- Converts **private IP** → **public IP**
- Allows multiple devices to access internet using one IP
- Provides basic security

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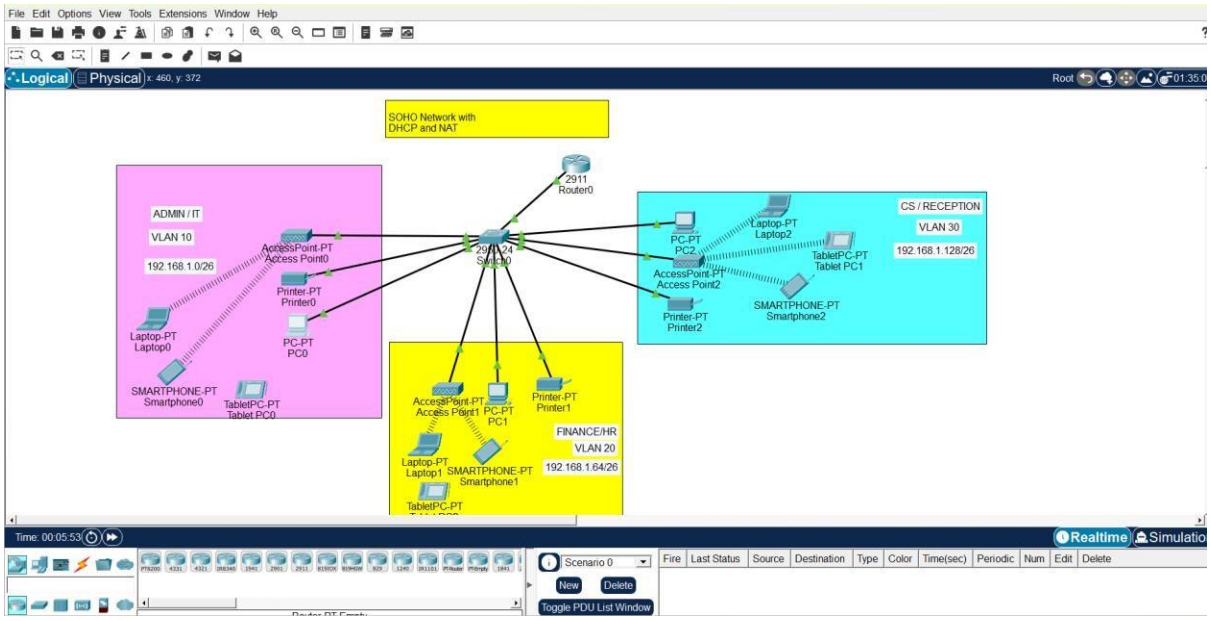
5. Network Diagram



6. IP Addressing Scheme

Device	IP Address	Subnet Mask
Router LAN	192.168.1.1	255.255.255.0
Router WAN	Public IP (e.g., 200.0.0.1)	-
PCs	DHCP Assigned	255.255.255.0

7. Configuration

◆ Router Configuration (LAN)

```
interface fa0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
```

◆ DHCP Configuration

```
ip dhcp pool LAN
network 192.168.1.0 255.255.255.0
default-router 192.168.1.1
```

◆ NAT Configuration

```
access-list 1 permit 192.168.1.0 0.0.0.255
ip nat inside source list 1 interface fa0/1 overload
```

◆ Interface Configuration

```
interface fa0/0
ip nat inside

interface fa0/1
ip nat outside
```

---

8. Procedure

1. Place router, switch, and PCs
2. Connect all devices
3. Configure router interfaces
4. Enable DHCP
5. Configure NAT
6. Test connectivity

G. Testing

◆ Internal Test

```
ping 192.168.1.x
```

◆ Internet Test

```
ping 8.8.8.8
```

— Successful replies = Network working

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10. Result

The SOHO network was successfully implemented with DHCP and NAT. Devices received IP automatically and accessed external networks.

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11. Advantages

- Automatic IP assignment
- Efficient internet sharing
- Secure internal network

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12. Limitations

- Depends on router performance
- Limited scalability

(Networking s Devices)

Parameter	Details
Network Type	SOHO Network
Connecting Devices	Router, switch, PCs, ISP cloud
Explanation	SOHO networks use DHCP for automatic IP allocation and NAT to allow multiple devices to access the internet securely using a single public IP.